Network Analysis via Wireshark

1. History (Wireshark ki Shuruat) :-

- Original Name: Wireshark was originally launched under the name Ethereal.
- Founder: It was started by Gerald Combs in 1998 to analyze network traffic and solve network problems.
- Renaming: Due to trademark issues, the project was renamed from Ethereal to Wireshark in 2006.
- Purpose: It is currently the world's most widely-used Network Protocol Analyzer.

2.Installation Section (How to Setup):-

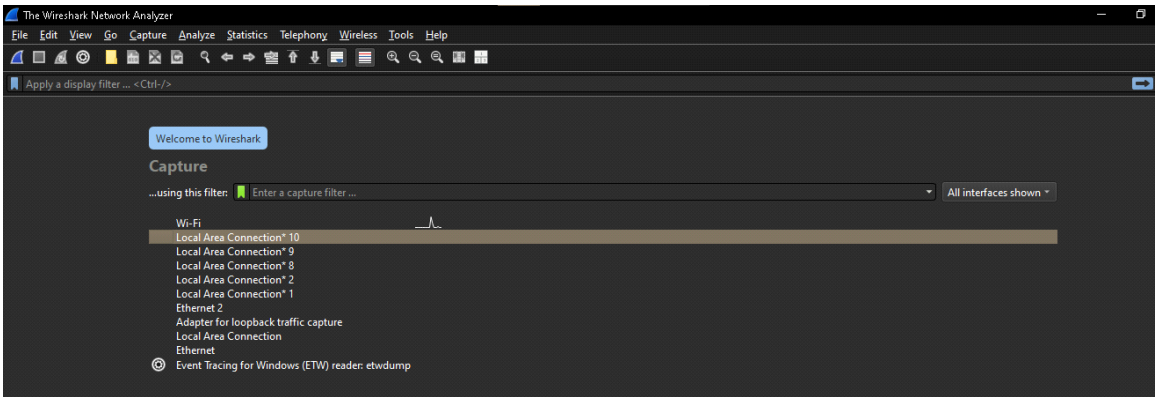
- Official Website- https://www.wireshark.org/download.html
- Steps: 1. Download -> 2. Install (with Npcap) -> 3. Run as Administrator.

3. Practical :-

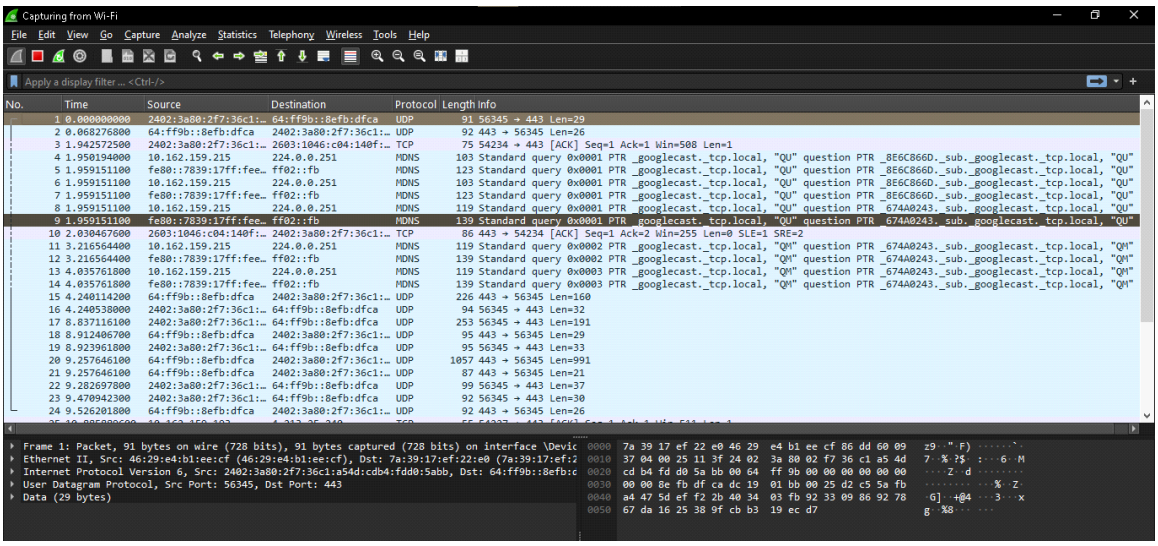
a. Wireshark interface -

i. capture filterb bare ,interface list

ii. interface list :- Multiple options are appearing, and the one with the slash (or line) through it is the active internet connection.



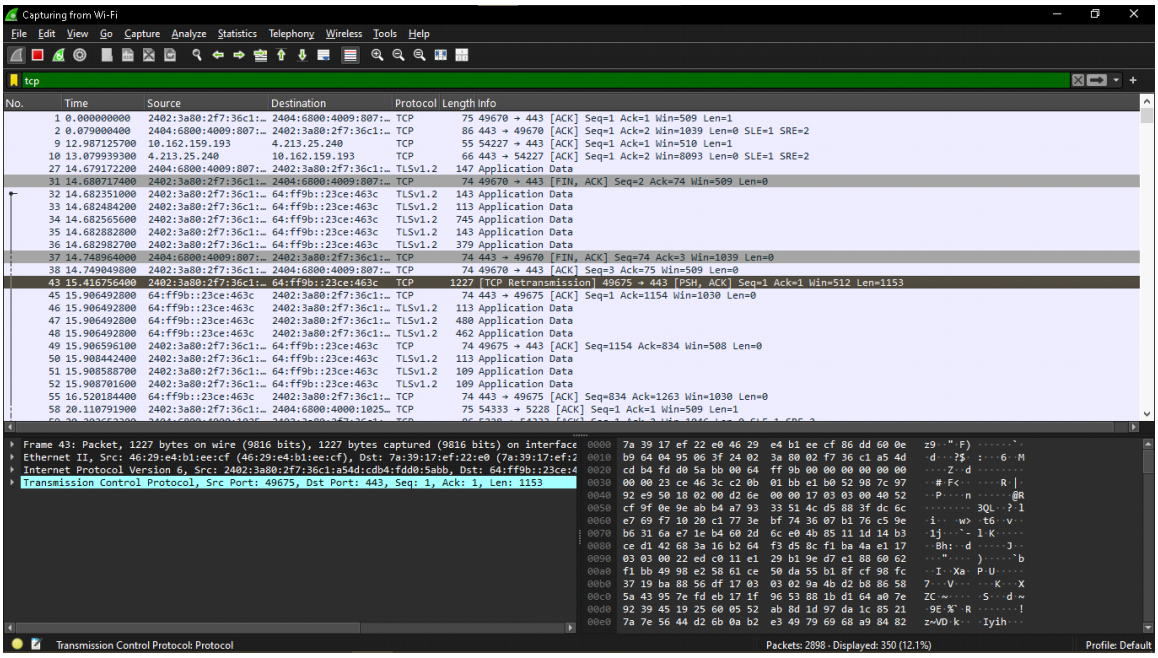
b. Post-capture screen :-



* **Sniffer :-** Wireshark does not create packets; instead, it captures and displays copies of network traffic for analysis.

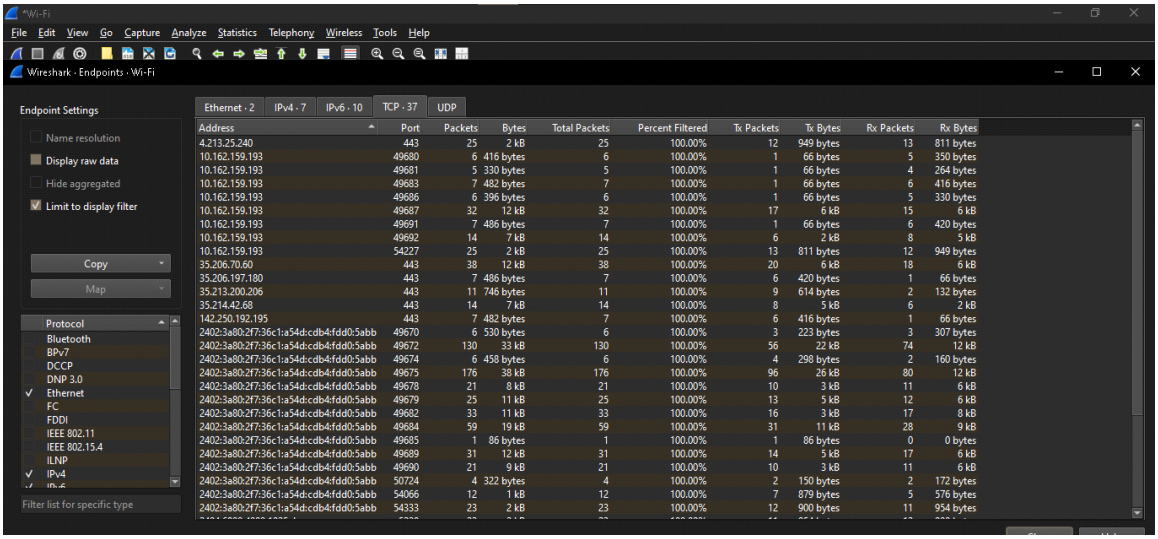
c. Applying Wireshark Filters :- Filters simplify network analysis by hiding 'noise' (unnecessary data) and displaying only the specific packets you want to see.

- TCP/TLS - TCP and TLS filters are used for deep traffic analysis and security auditing.



3. Monitoring Network Traffic & Data Usage :- In the IP Address and Protocol filters, we check which IP addresses we have visited, how many bytes were used, and whether the connection is using IPv4 or IPv6.

* We don't open the packets; we only see who is using the most internet. This is a key strategy for network administrators.

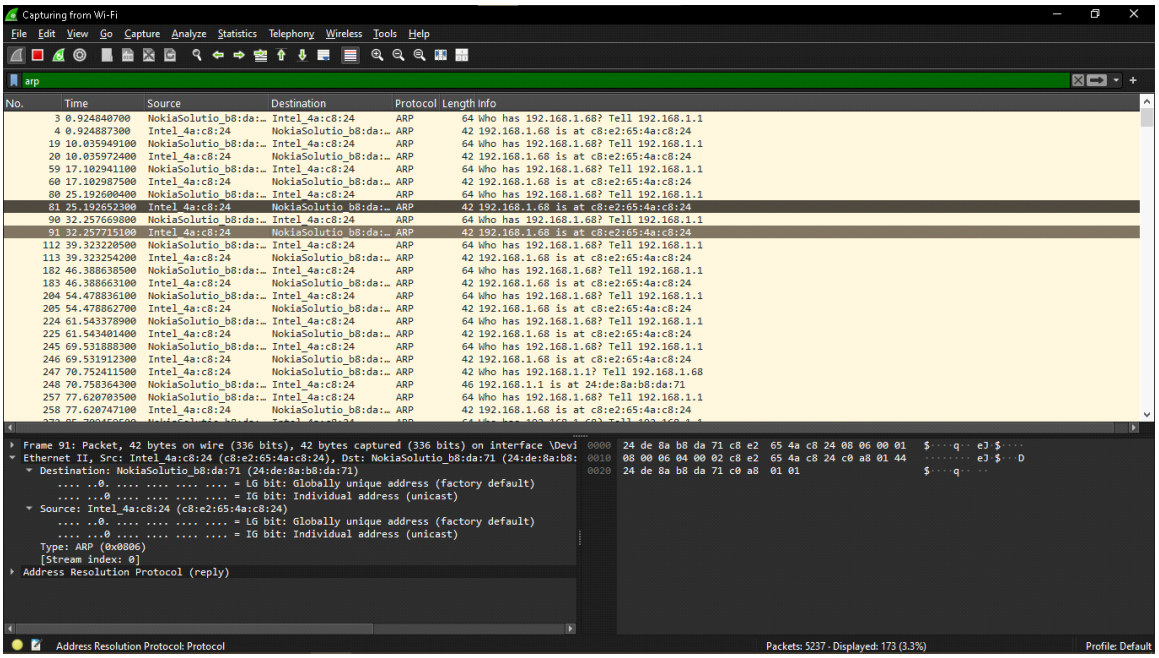


4.**Network(ARP Protocol)** :-

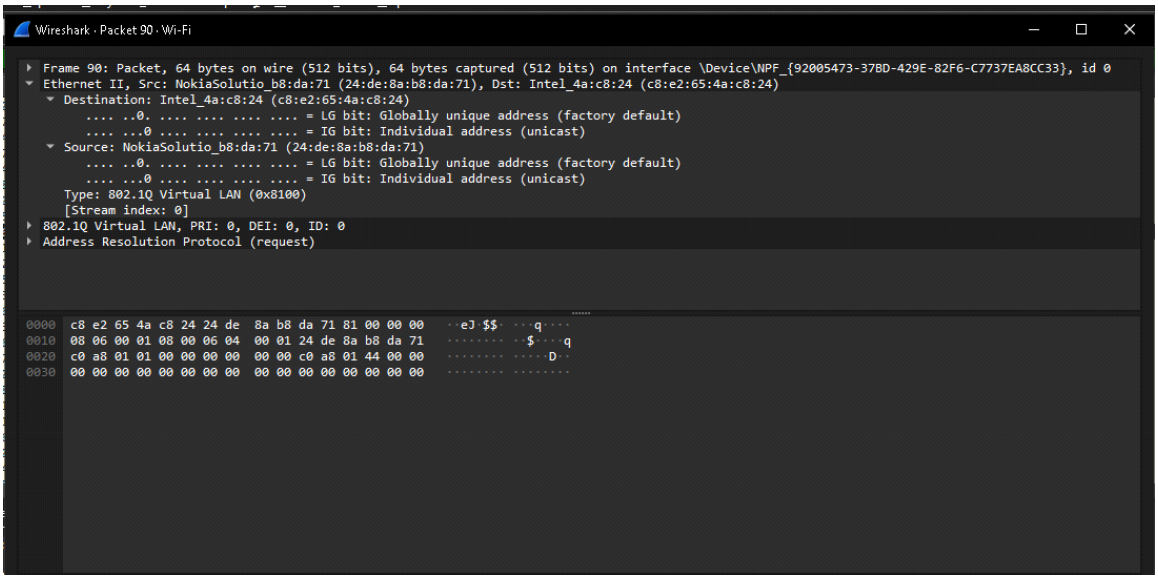
The primary purpose of ARP is to facilitate PC-to-PC communication within a local network. While computers use IP addresses to identify each other globally, they need MAC addresses (Physical Addresses) to actually deliver data packets over a local Ethernet or Wi-Fi connection.

* A computer’s real identity is its MAC address, not its IP. While an IP can change, the MAC address is a permanent hardware ID hardcoded into the network card by the manufacture

1. ARP filter use:-

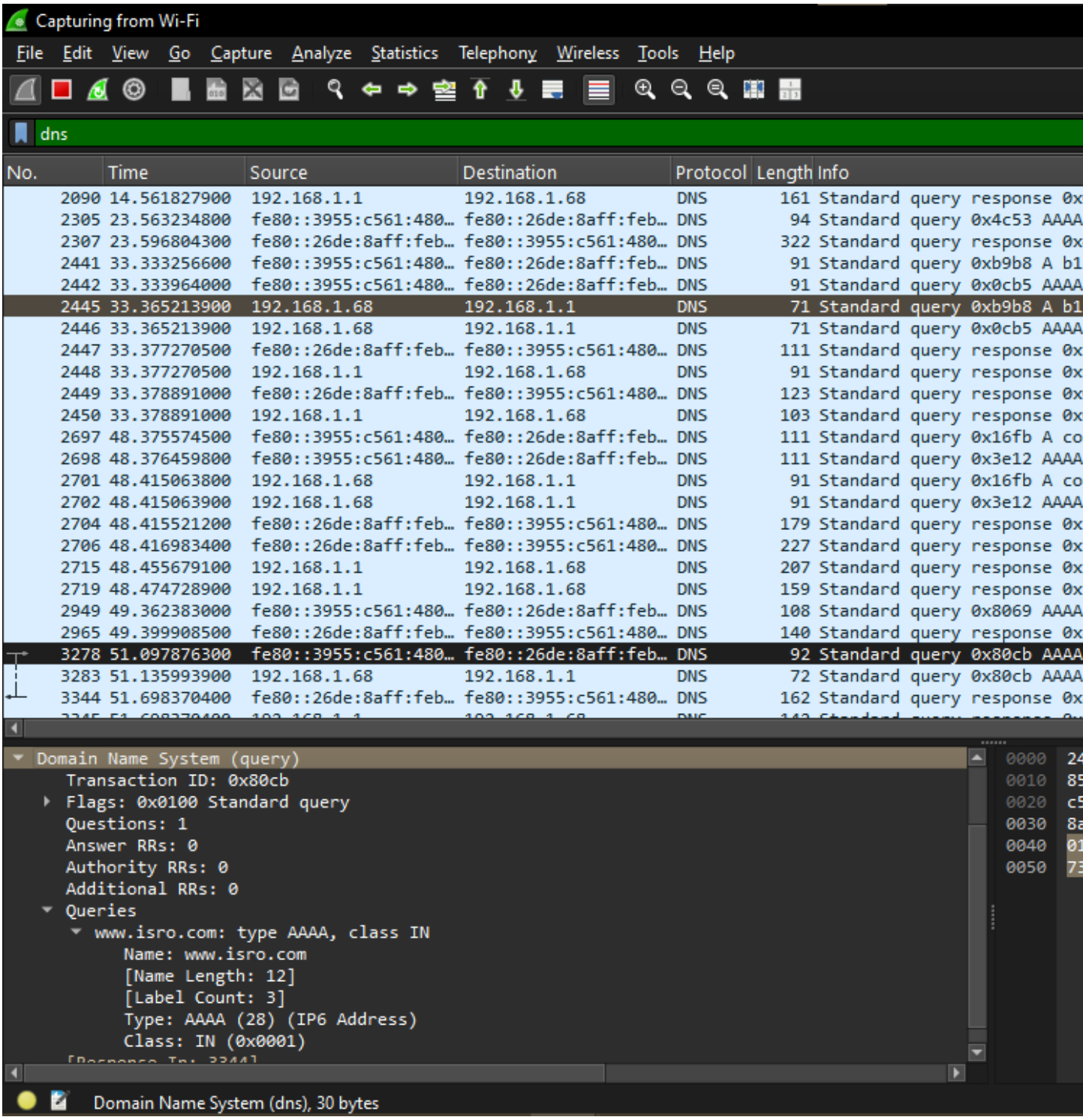


2. AIR through MAC addresh see:



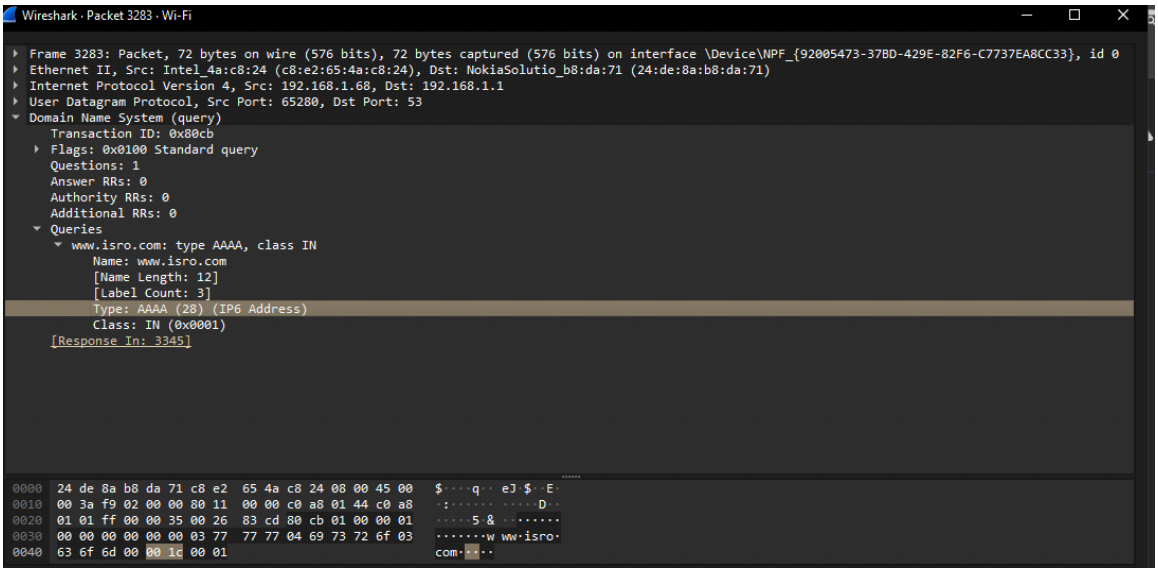
5. DNS packet analysis :-

- Step 1: Capture & Filter
- Open Wireshark: Select your active Wi-Fi/Ethernet and click the blue shark fin to start.
- Search Site: Open a browser, go to any website (e.g.www.isro.com,), then return to Wireshark.
- Apply Filter: Type dns in the top filter bar and press Enter to show only DNS traffic.
- Find Packet: Look for the "Standard query" line containing your website name and click on it.-



- Step 2: Deep Analysis

- Check Transaction ID: In the middle pane, expand "Domain Name System" to see the Transaction ID (matches queries to responses).
- View IP Address: Expand the "Answers" section in a "Response" packet to find the website’s actual IP Address.
- Identify Type: Verify the Query Type (e.g., A for IPv4 or AAAA for IPv6) to confirm the connection type.



6. DHCP -D.O.R.A :- DHCP is a service that automatically gives an IP address to any device that joins a network, saving you from manual setup.

1.The DORA Process :-

- When a device connects, it follows these 4 steps:
- Discover: Device asks, "Is there a Server?"
- Offer: Server says, "Yes, here is an IP for you."
- Request: Device says, "Okay, I'll take it!"
- Acknowledge: Server says, "Done! You are now connected."

Lab: Observing DORA Process in Wireshark

i. Open & Start: Open Wireshark → Select Wi-Fi/Ethernet → Click Blue Fin (Start).

ii. Filter: Type dhcp in the filter bar and press Enter.

iii. Reset Wi-Fi: Turn your Wi-Fi OFF then ON to force a new connection.

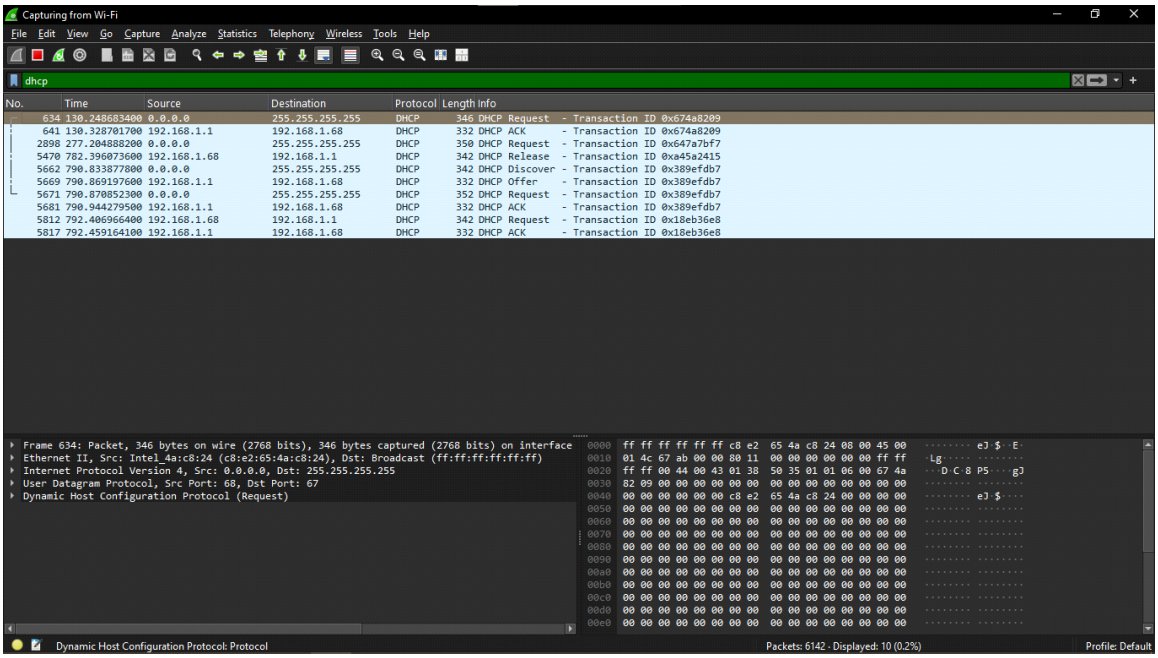
iv. Observe DORA: Look for these 4 packets:

v. Discover: PC searching for server.

vi. Offer: Router offering IP.

vii. Request: PC asking for that IP.

viii. ACK: Router confirming the deal.



7. HTTP Practical Analysis (Wireshark):- To capture and analyze unencrypted HTTP traffic to understand how the Client (Browser) and Server communicate in plain text.

Quick Summary: HTTP Practical in Wireshark

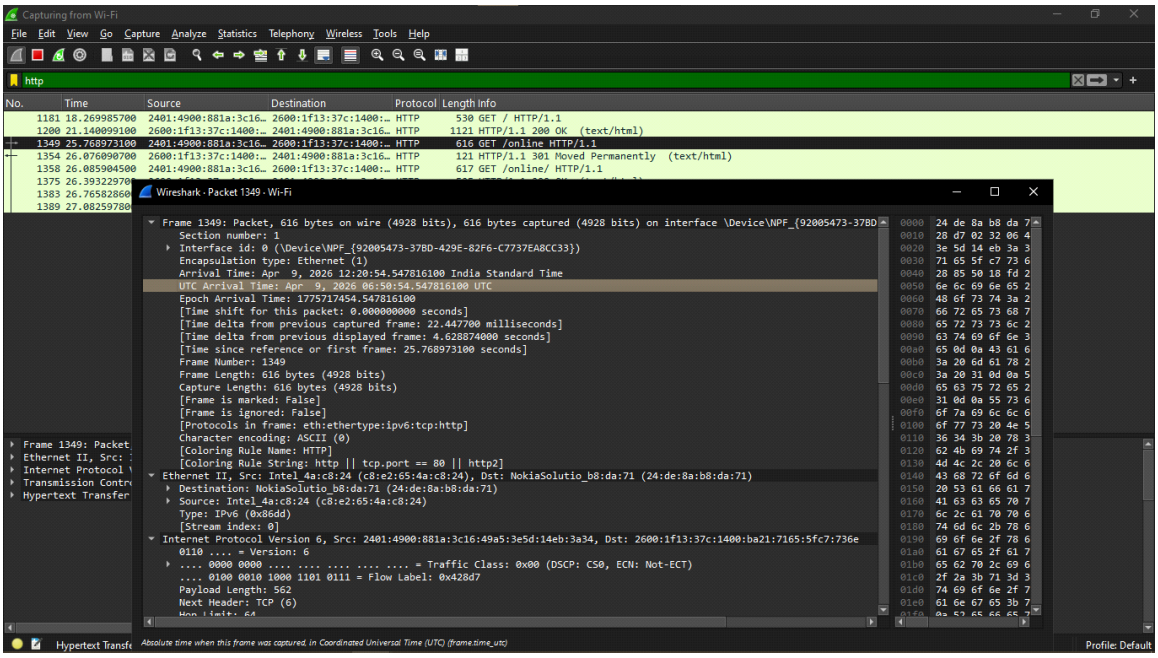
a. Step 1 (Start): Open Wireshark -> Select the Network Interface -> Click the Blue Shark icon to start the capture.

b. Step 2 (Filter): Type http in the top filter bar and press Enter (to see only web data).

c. Step 3 (Action): Go to your browser and search for http://neverssl.com.

d. Step 4 (Stop): Return to Wireshark and click the Red Square button to stop the capture.

e. Step 5 (Analysis): Find the packet with "GET / HTTP/1.1" and check the details below.



8.**Analyzing ICMP Traffic (Ping) in Wireshark:-** ICMP is a network layer protocol used to send error messages and check if a host or server is reachable (Ping).

ICMP Analysis (Ping Test)

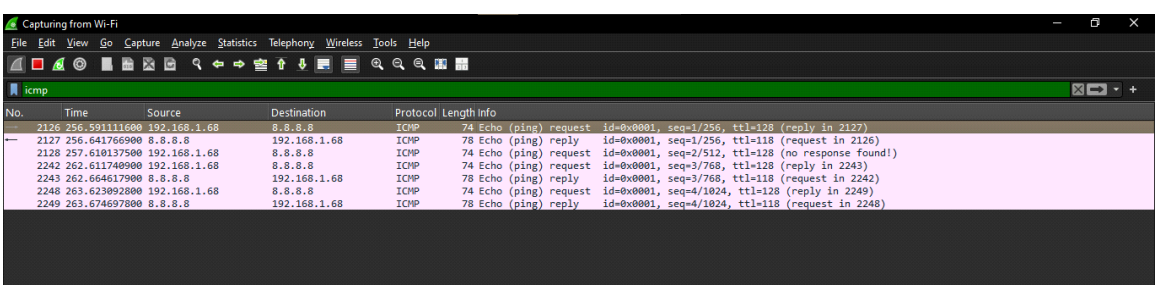
i. Step 1 (Capture): Open Wireshark -> Select your Interface -> Click the Blue Shark icon.

ii. Step 2 (Filter): Type icmp in the display filter bar and press Enter.

iii. Step 3 (Action): Open CMD, type ping 8.8.8.8, and hit Enter.

iv. Step 4 (Stop): Go back to Wireshark and click the Red Square button.

v. Step 5 (Analysis): Select the Echo Request/Reply packets to analyze the Layer 3 details.

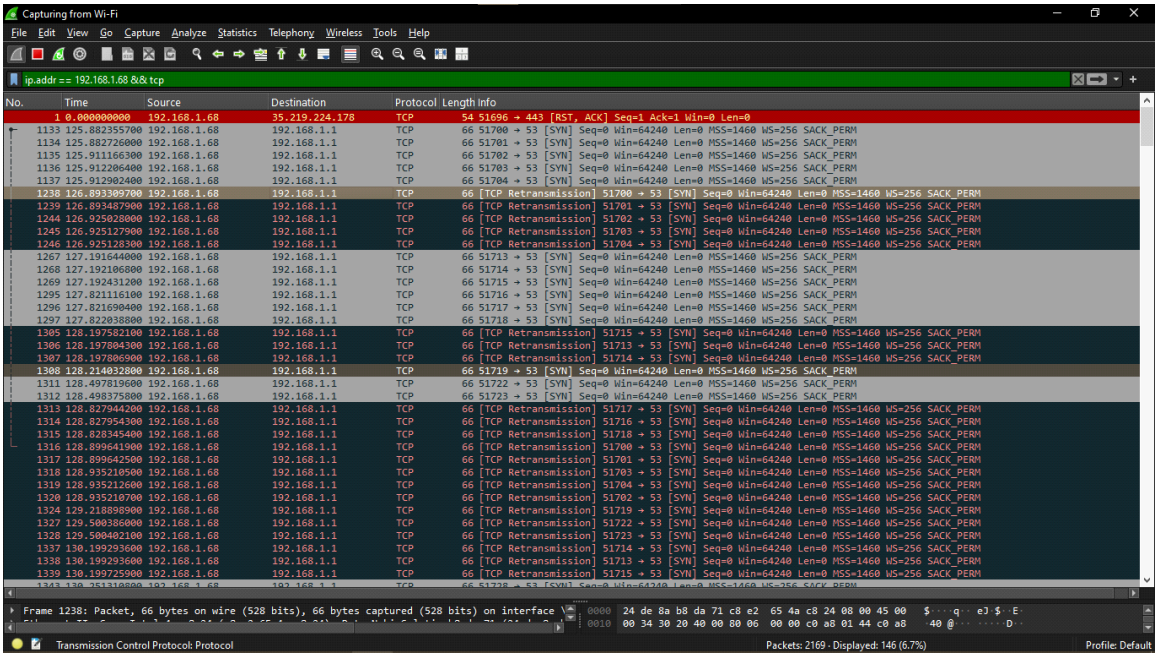


8.Advanced Display Filters:-

*** when using multiple files in Wireshark, you merge them first and then use Logical Operators**

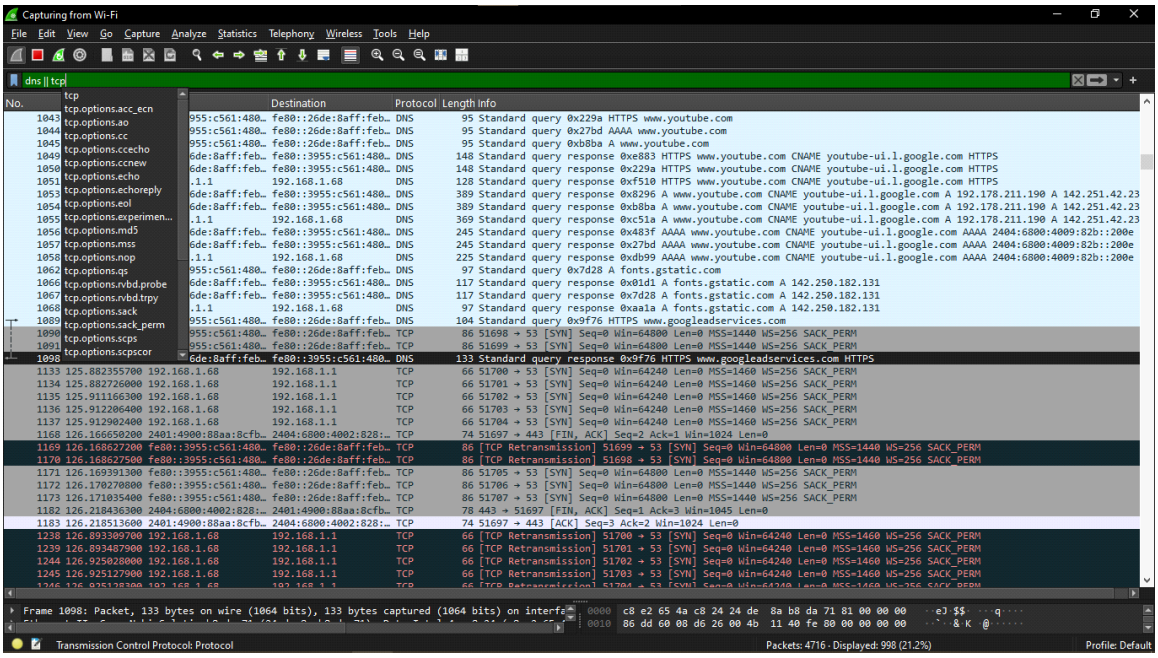
to filter the data.

1. ip.addr<ip> && http :- Yes, precisely. Using ip.addr == <IP_ADDRESS> && http is the standard way to isolate HTTP traffic for a specific device in Wireshark.

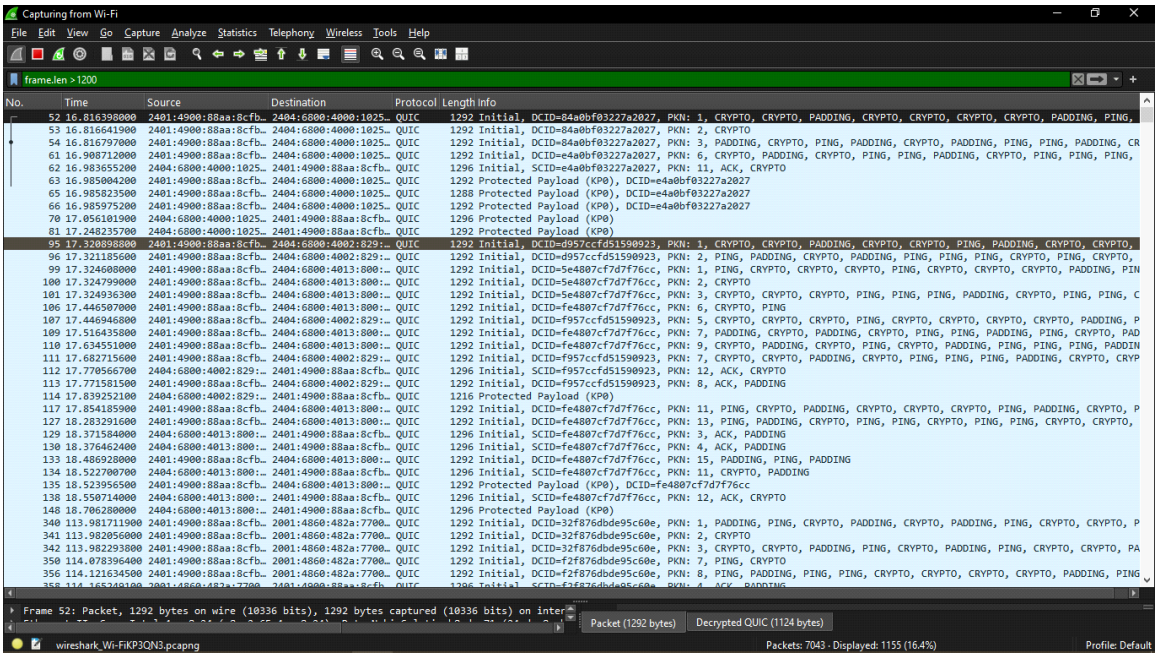


2. DNS || TCP :- The OR operator tells Wireshark to show packets that match either of the protocols you listed. It is used when you want to see multiple types of traffic in the same list at the same time.

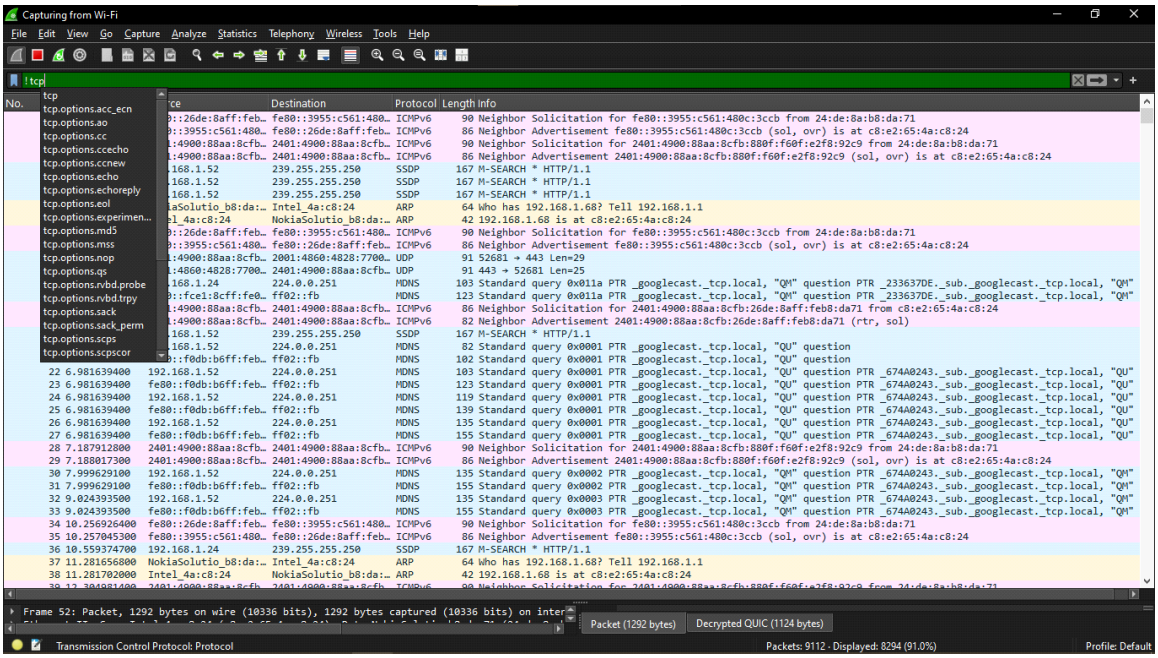
* This filter Using all dns and tcp filter one time same page show.



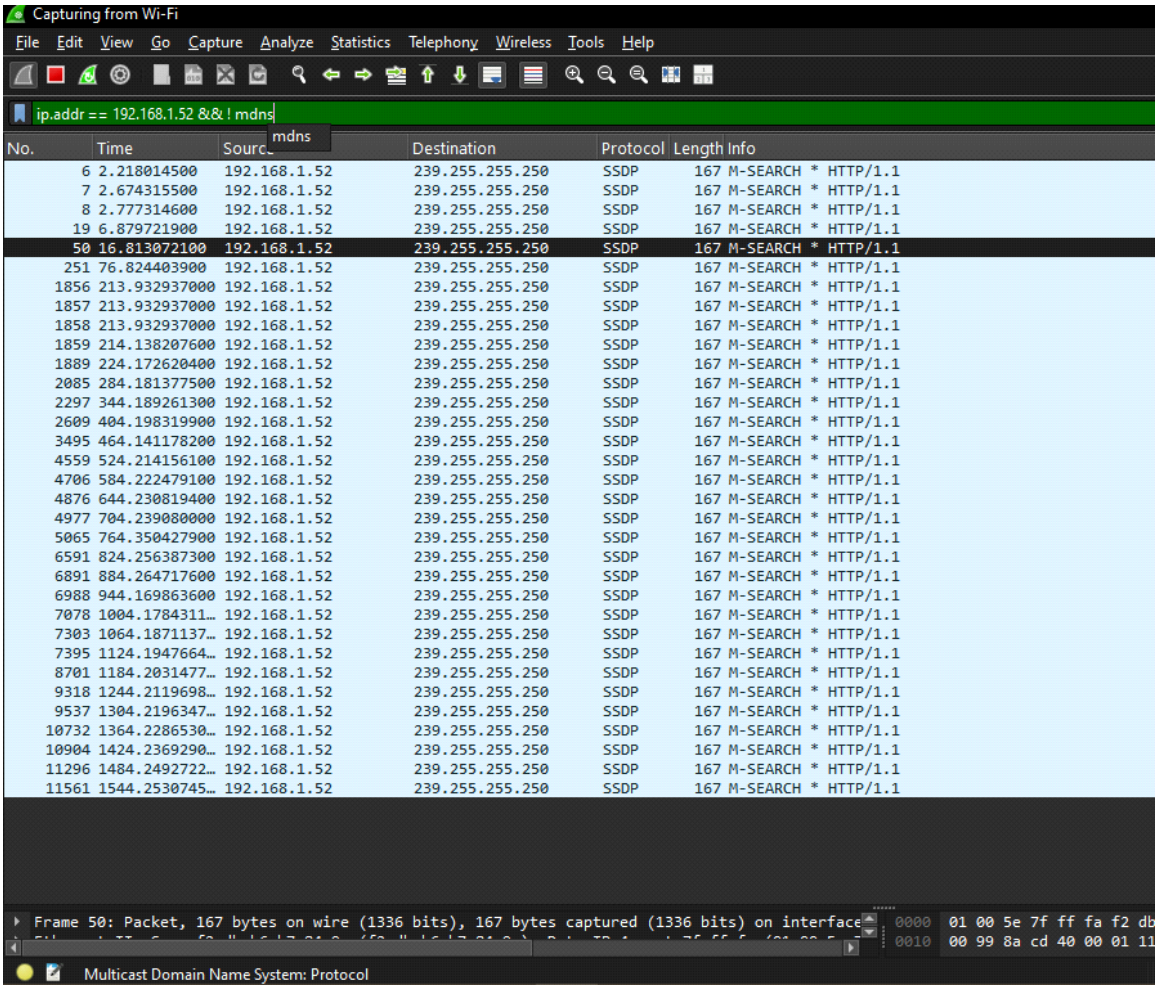
3. Frame.len > 1200 :- The filter frame.len > 1200 in Wireshark is used to find packets based on their physical size on the wire.



4. **! tcp :-** acts as a block-list for TCP, showing you everything else that is happening on the network.

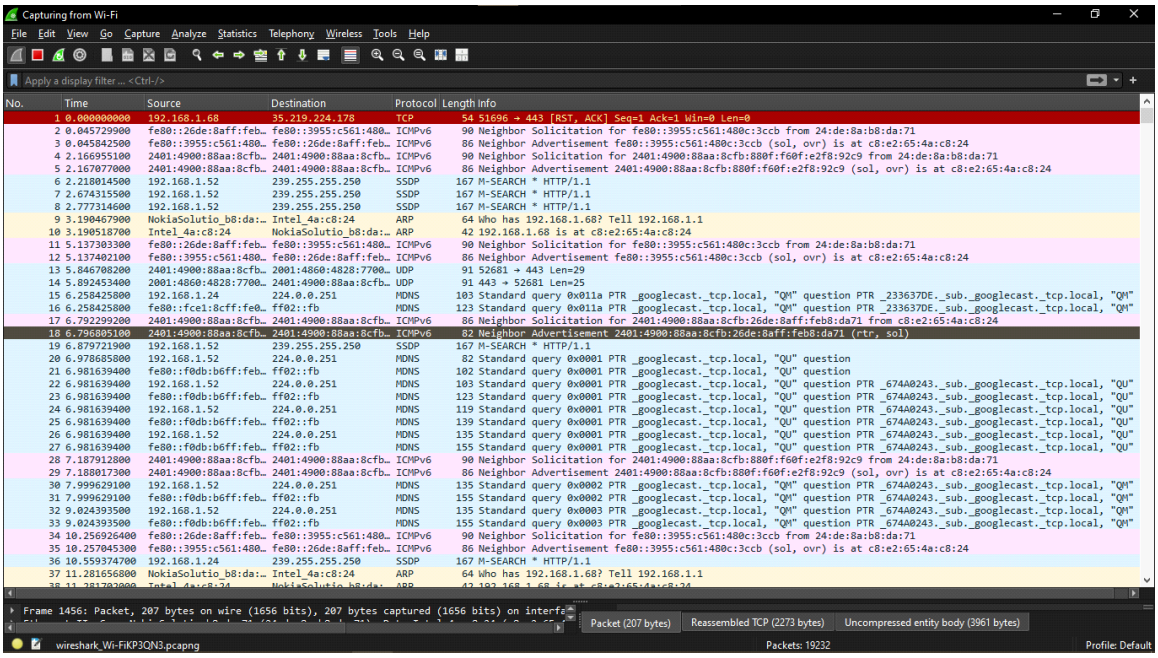


5. ip.addr == <ip> && ! mdns :- Show me everything related to this specific IP, but filter out all the mDNS packets from the results.



6.Wireshark Coloring Rules:- In Wireshark, the different colors represent Coloring Rules that help you identify protocols and potential network issues at a glance.

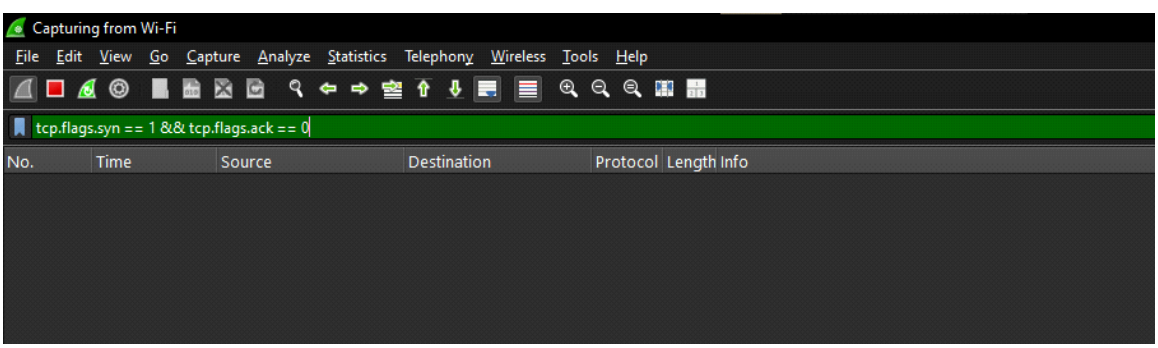
- Green: HTTP traffic.
- Dark Blue: DNS.
- Light Blue: UDP.
- Black/Red: Inme koi Error ya Problem hoti hai.



10.Network security - Attack Ifentification :-

1. port scan :- If you notice a single IP address sending a massive amount of SYN packets to various ports (like 21, 22, 23, 443, etc.) in your Wireshark capture, it is a clear sign of a Port Scan or a SYN Flood attack.

filter :- tcp.flags.syn ==1 && tcp.flags.ack == 0



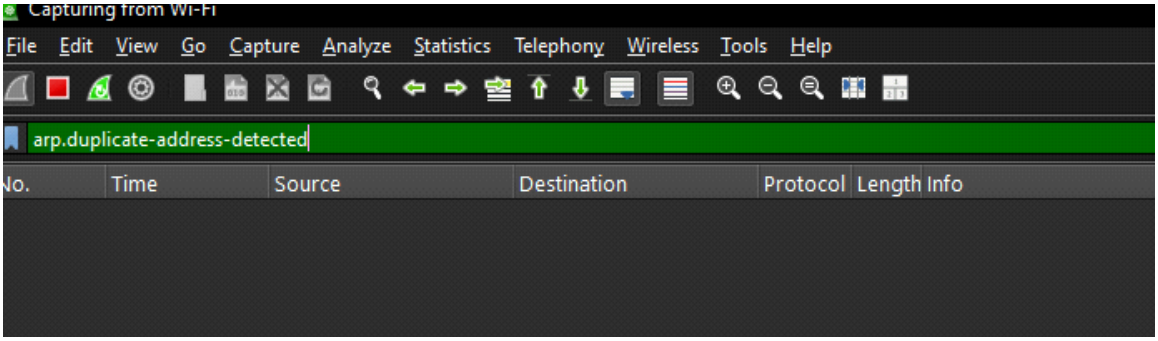
2. Dos /flood Attack :- A DoS attack is a malicious attempt to disrupt the normal traffic of a targeted server, service, or network by overwhelming the target or its surrounding

infrastructure with a flood of Internet traffic.

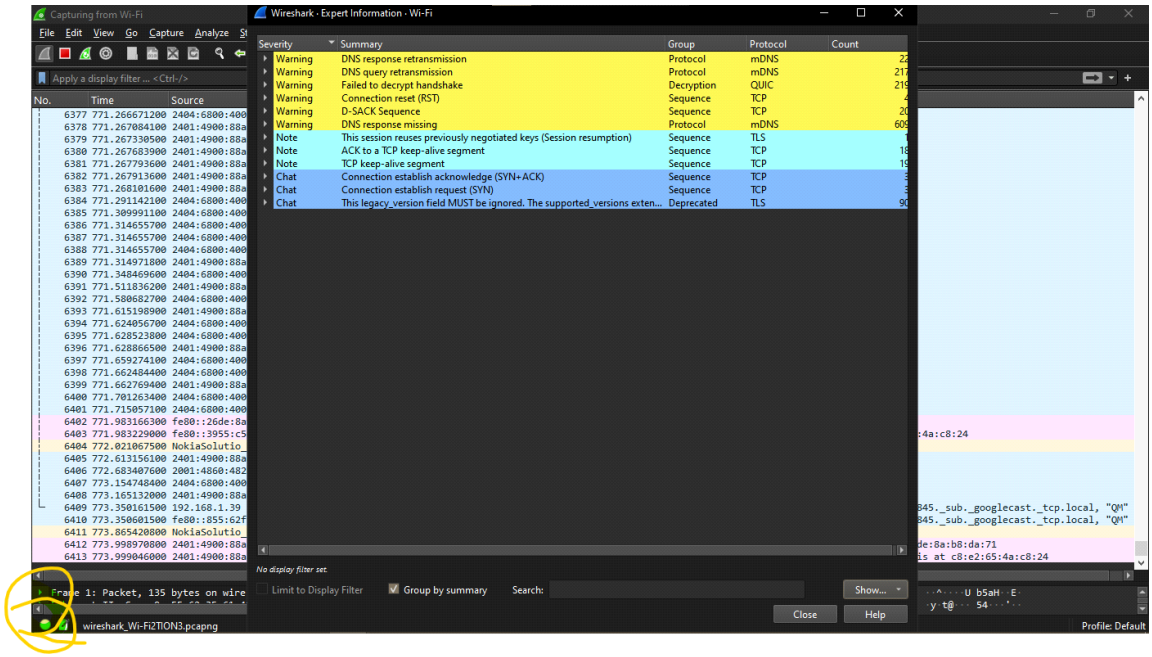
Selustion :- red/blak color, packts show(Attack for hackers)

3. ARO spoofing :- ARP Spoofing is a type of cyberattack where an attacker sends falsified ARP (Address Resolution Protocol) messages over a local area network (LAN). The goal is to link the attacker's MAC address with the IP address of a legitimate device (like the Default Gateway or Router).

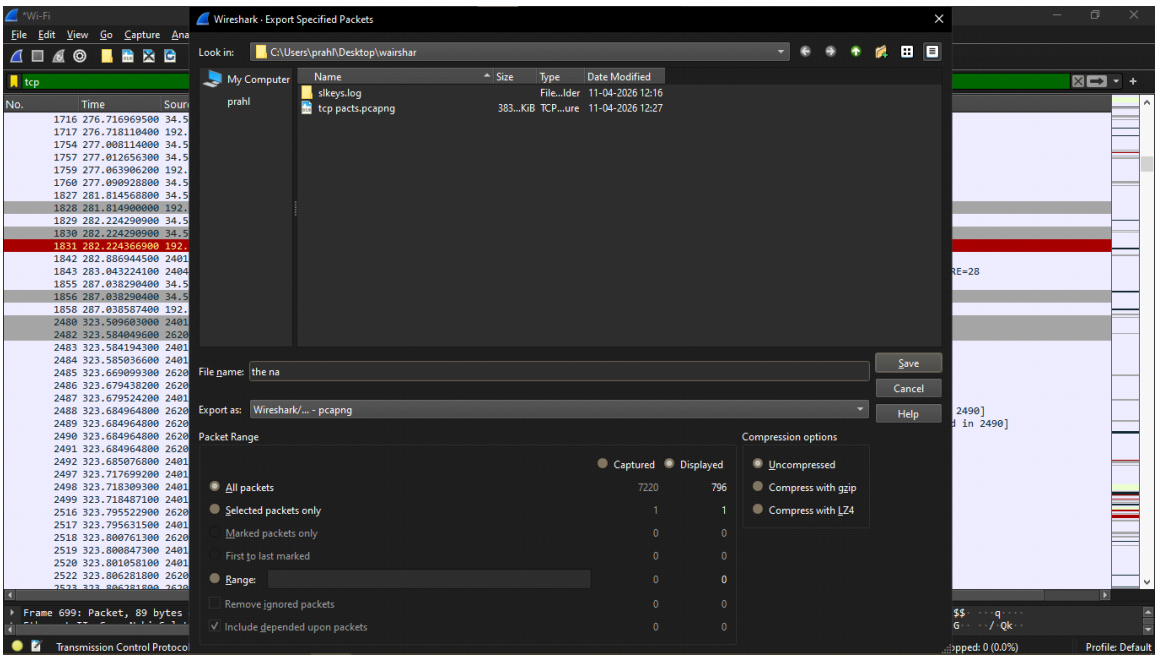
filter :- arp.duplicate-address-detected



4. EXPERT INFORMASTION :-he Expert Information tool in Wireshark is a built-in diagnostic feature that automatically analyzes captured network traffic to identify anomalies, errors, and performance issues. It categorizes packets based on their behavior and provides a summary of potential network problems without requiring manual inspection of every individual packet.



11. Pro - laval Reporting :-

1. Traffic save:-

- Filter: Apply a display filter (e.g., http or ip.addr == x.x.x.x) to show only the relevant traffic.
- Navigate: Go to File > Export Specified Packets...
- Selection: In the "Packet Range" section, ensure "Displayed" is selected (this ignores the background noise and saves only your filtered results).
- Format: Choose .pcapng as the file type (the industry standard for compatibility and metadata).
- Save: Name your file and click Save.
- Would you like to know the most common display filter